

RASHEEQUE AHAMMED RAHIM

AI/ML Engineer | Model Development & LLM Ops | Debugging & Research | Production ML

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PROFESSIONAL SUMMARY

Problem-driven AI/ML Engineer specializing in model development, debugging, and productionization. I design, refine, and deploy predictive and generative models, investigate root causes of model failures, and implement robust retraining and monitoring workflows. Experienced with LLM operations and guardrails, RAG-based retrieval, and building evaluation/validation harnesses for safe, high-quality outputs. Comfortable translating research into practical systems and collaborating cross-functionally to ship reliable AI products.

TECHNICAL SKILLS

Model Development & Debugging: Model diagnostics, root-cause analysis, performance profiling, monitoring, unit tests, A/B testing, evaluation metrics

LLM Operations: Prompt engineering, RAG pipelines, guardrails, safety checks, evaluation harnesses, retraining workflows, LangChain, LangGraph

ML Frameworks: PyTorch, TensorFlow, Scikit-Learn, Keras, model training, evaluation, hyperparameter tuning

Computer Vision: OpenCV, object detection (YOLO), image segmentation, custom dataset training, data augmentation

Programming & Data: Python, Pandas, NumPy, SQL, ETL, feature engineering, statistical analysis

Deployment & MLOps: FastAPI, REST APIs, Docker, Kubernetes basics, CI/CD, model versioning, cloud platforms (AWS, GCP, Azure)

PROFESSIONAL EXPERIENCE

AI ML Development Engineer

May 2025 – Present

Think Neural AI

Hybrid

- Designed, tuned, and deployed predictive and generative models; conducted detailed model diagnostics and root-cause analysis to resolve failures and improve reliability.
- Implemented LLM operations and RAG pipelines with guardrails and evaluation harnesses; designed retraining triggers and model monitoring for continuous improvement.
- Built API-driven AI/ML services using FastAPI and containerized deployments with Docker, collaborating with engineering teams on CI/CD integration, testing, and performance optimization for production readiness. Deployed AI systems on AWS cloud infrastructure

Data Scientist Intern

March 2024 – October 2024

Innodatatics

Bangalore, India

- Developed and debugged NLP models, performing error analysis and iterative improvements through feature engineering and statistical validation.
- Applied supervised and unsupervised algorithms, supported ETL and preprocessing pipelines, and produced clear experiment reports to communicate model behavior and recurring issues.

KEY PROJECTS

Sakshi.AI - Real-Time Video Analytics Platform

May 2025 – Present

- Built a real-time computer vision system using YOLO, NVIDIA DeepStream, and custom segmentation models to enable intelligent video analytics across 100+ camera feeds with PyTorch.
- Deployed production-ready inference service using FastAPI with Docker containerization; optimized model latency and accuracy for real-time edge deployment scenarios. *Tech: PyTorch, YOLO, Deepstream, OpenCV, FastAPI, Docker, Python*

Generative AI Assistant - RAG & LLM Fine-tuning

November 2024 – January 2025

- Designed and implemented an LLM use case with production guardrails: fine-tuning, RAG retrieval, safety filters, and an automated evaluation harness to measure output quality under realistic scenarios.
- Established retraining pipelines and monitoring metrics to detect drift and trigger model refresh; documented lessons learned and literature review insights to guide model improvements. *Tech: PyTorch, LangChain, Hugging Face, ChromaDB, FastAPI*

SpaceZap.AI - Spatial AI & Generative Design System

May 2025 – Present

- Developed a generative AI system combining computer vision (3D point cloud processing) and deep learning models for intelligent spatial design generation across 50+ retail scenarios.
- Engineered API service using FastAPI, containerized with Docker for cloud deployment; integrated model serving, validation pipelines, and scalable inference. *Tech: PyTorch, OpenCV, FastAPI, Docker, Python, SQL*

EDUCATION

Masters in Computer Application (MCA) - AI Specialization

2023 – 2025

Amrita Vishwa Vidyapeetham (Amrita AHEAD) - Focus: ML, GenAI, Deep Learning, NLP

Bachelor of Science in Computer Science

2020 – 2023

College of Applied Science Koothuparamba, Kannur, India